

COMP4121 Lecture Notes

Linear Programming

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We now move to one of the most important cases of convex programming, called *Linear Programming*, (*LP*), in which the objective is a linear function and the convex domain over which the extremum is sought is defined by constraints which are also linear functions. We follow closely our old *COMP3121* textbook by Cormen, Leiserson, Rivest and Stein, *introduction to Algorithms*. We start by defining a common representations of Linear Programming optimization problems.

In the **standard form** the *objective* to be maximized is given by

$$f(\mathbf{x}) = \sum_{j=1}^n c_j x_j,$$

and the *constraints* are of the form

$$\sum_{j=1}^n a_{ij} x_j \leq b_i, \quad 1 \leq i \leq m; \quad (1)$$

$$x_j \geq 0, \quad 1 \leq j \leq n, \quad (2)$$

Note that $\mathbf{x}$ represents a vector, $\mathbf{x}^\top = \langle x_1, \dots, x_n \rangle$; we assume that vectors are columns; thus, to write them as rows we have to transpose them. The numbers a_{ij} are assumed to be real.

Clearly, a minimization problem of the form:

$$\text{minimize } f(\mathbf{x}) = \sum_{j=1}^n c_j x_j,$$

subject to the constraints of the form

$$\sum_{j=1}^n a_{ij} x_j \geq b_i, \quad 1 \leq i \leq m; \quad (3)$$

$$x_j \geq 0, \quad 1 \leq j \leq n, \quad (4)$$

can easily be reduced to a corresponding maximization problem in the standard form

$$\text{maximize } f^*(\mathbf{x}) = \sum_{j=1}^n c_j^* x_j,$$

with constraints of the form

$$\sum_{j=1}^n a_{ij}^* x_j \leq b_i^*, \quad 1 \leq i \leq m;$$

$$x_j \geq 0, \quad 1 \leq j \leq n,$$

by taking $\mathbf{c}^* = -\mathbf{c}$, $a_{ij}^* = -a_{ij}$ and $\mathbf{b}^* = -\mathbf{b}$.

Example: Humans require 13 vitamins $V_1, \dots, V_{13}$ in total; the daily required amounts $d_1, \dots, d_{13}$ for each of them are known. Your local grocery store carries 130 types of food staples, and for each food staple S_k and each of the 13 vitamins V_i the content of V_i per gram of S_k is also a known value, denoted by $C(k, i)$. For each food staple S_k you are also given its price per gram, denoted by P_k . Your goal is to determine the quantities x_k of each food staple S_k which you should buy, such that your daily requirements of all vitamins are met, but the total price you pay for your daily food provision is as small as possible.

Solution: You have to:

$$\begin{aligned} & \text{minimize the objective} && \sum_{k=1}^{130} P_k x_k \\ & \text{subject to 13 constraints} && \sum_{k=1}^{130} C(k, i) x_k > d_i, \quad 1 \leq i \leq 13, \\ & && x_1 \geq 0, \dots, x_{130} \geq 0. \end{aligned}$$

Somewhat messy formulation of an LP program we have given can be reformulated in a much more compact matrix form. To get a very compact representation of linear programs, let us introduce a partial ordering on vectors $\mathbf{x} \in \mathbb{R}^n$ by $\mathbf{x} \leq \mathbf{y}$ if and only if such inequalities hold coordinate-wise, i.e., if and only if $x_j \leq y_j$ for all $1 \leq j \leq n$. Since vectors are written as a single column matrices, letting $\mathbf{c}^\top = \langle c_1, \dots, c_n \rangle \in \mathbb{R}^n$ and $\mathbf{b}^\top = \langle b_1, \dots, b_m \rangle \in \mathbb{R}^m$, and letting $\mathbf{A}$ be the matrix $A = (a_{ij})_{i,j}$ of size $m \times n$, we get that the above problem can be formulated simply as:

$$\text{maximize } \mathbf{c}^\top \mathbf{x}$$

subject to the following two (matrix-vector) constraints:¹ $\mathbf{Ax} \leq \mathbf{b}$ and $\mathbf{x} \geq \mathbf{0}$.

Thus, to specify a Linear Programming optimisation problem we just have to provide a triplet $(\mathbf{A}, \mathbf{b}, \mathbf{c})$, and the information that the triplet represents an LP problem in the standard form.

While in general the “natural formulation” of a problem into a Linear Program does not necessarily produce the non-negativity constraints (4) for all of the variables, in the standard form such constraints are indeed required for all of the variables. However, this poses no problem, because each occurrence of an unconstrained variable x_j can be replaced by the expression $x'_j - x^*_j$ where x'_j, x^*_j are new variables satisfying the constraints $x'_j \geq 0, x^*_j \geq 0$.

Any vector $\mathbf{x}$ which satisfies the two constraints is called a *feasible solution*, regardless of what the corresponding objective value $\mathbf{c}^\top \mathbf{x}$ might be. Note that the set of feasible solutions, i.e., the domain over which we seek to maximize the

¹Note that the inequality involved in the constraints is the partial ordering on vectors which we have just introduced above. Also, in the non-negativity constraint, $\mathbf{0}$ represents a vector $\mathbf{0} \in \mathbb{R}^n$ with all coordinates equal to 0.

objective, is a convex set because it is an intersection of the half spaces defined by each of the constraints.

As an example, let us consider the following optimization problem:

$$\begin{array}{ll} \text{maximize} & 3x_1 + x_2 + 2x_3 \end{array} \quad (5)$$

subject to the constraints

$$x_1 + x_2 + 3x_3 \leq 30 \quad (6)$$

$$2x_1 + 2x_2 + 5x_3 \leq 24 \quad (7)$$

$$4x_1 + x_2 + 2x_3 \leq 36 \quad (8)$$

$$x_1, x_2, x_3 \geq 0 \quad (9)$$

How large can the value of the objective $z(x_1, x_2, x_3) = 3x_1 + x_2 + 2x_3$ be, without violating the constraints? If we add inequalities (6) and (7), we get

$$3x_1 + 3x_2 + 8x_3 \leq 54$$

Since all variables are constrained to be non-negative, we are assured that

$$3x_1 + x_2 + 2x_3 \leq 3x_1 + 3x_2 + 8x_3 \leq 54$$

The far left hand side of this equation is just the objective (5); thus $z(x_1, x_2, x_3)$ is bounded above by 54, i.e., $z(x_1, x_2, x_3) \leq 54$.

Can we obtain a tighter bound? We could try to look for coefficients $y_1, y_2, y_3 \geq 0$ to be used to for a linear combination of the constraints:

$$y_1(x_1 + x_2 + 3x_3) \leq 30y_1$$

$$y_2(2x_1 + 2x_2 + 5x_3) \leq 24y_2$$

$$y_3(4x_1 + x_2 + 2x_3) \leq 36y_3$$

Then, summing up all these inequalities and factoring, we get

$$\begin{aligned} x_1(y_1 + 2y_2 + 4y_3) + x_2(y_1 + 2y_2 + y_3) + x_3(3y_1 + 5y_2 + 2y_3) \\ \leq 30y_1 + 24y_2 + 36y_3 \end{aligned} \quad (10)$$

If we compare this with our objective (5) we see that if we choose y_1, y_2 and y_3 so that:

$$y_1 + 2y_2 + 4y_3 \geq 3$$

$$y_1 + 2y_2 + y_3 \geq 1$$

$$3y_1 + 5y_2 + 2y_3 \geq 2$$

then

$$x_1(y_1 + 2y_2 + 4y_3) + x_2(y_1 + 2y_2 + y_3) + x_3(3y_1 + 5y_2 + 2y_3) \geq 3x_1 + x_2 + 2x_3$$

Combining this with (10) we get:

$$30y_1 + 24y_2 + 36y_3 \geq 3x_1 + x_2 + 2x_3 = z(x_1, x_2, x_3)$$

Consequently, in order to find an as tight as possible upper bound for our objective $z(x_1, x_2, x_3)$, we have to look for y_1, y_2, y_3 which produce the smallest possible value of $z^*(y_1, y_2, y_3) = 30y_1 + 24y_2 + 36y_3$, but which do not violate the constraints

$$y_1 + 2y_2 + 4y_3 \geq 3 \quad (11)$$

$$y_1 + 2y_2 + y_3 \geq 1 \quad (12)$$

$$3y_1 + 5y_2 + 2y_3 \geq 2 \quad (13)$$

$$y_1, y_2, y_3 \geq 0 \quad (14)$$

Thus, trying to find the best upper bound for our objective $z(x_1, x_2, x_3)$ obtained by forming a linear combination of the constraints only reduces the original maximization problem to a minimization problem:

$$\begin{array}{ll} \text{minimize} & 30y_1 + 24y_2 + 36y_3 \\ \text{subject to the constraints} & (11-14) \end{array}$$

Such minimization problem P^* is called *the dual problem* of the initial problem P .

Let us now repeat the whole procedure with P^* in place of P , i.e., let us find the dual program $(P^*)^*$ of P^* . We are now looking for non negative multipliers $z_1, z_2, z_3 \geq 0$ to multiply inequalities (11-13) and obtain

$$z_1(y_1 + 2y_2 + 4y_3) \geq 3z_1$$

$$z_2(y_1 + 2y_2 + y_3) \geq z_2$$

$$z_3(3y_1 + 5y_2 + 2y_3) \geq 2z_3$$

Summing these up and factoring produces

$$y_1(z_1 + z_2 + 3z_3) + y_2(2z_1 + 2z_2 + 5z_3) + y_3(4z_1 + z_2 + 2z_3) \geq 3z_1 + z_2 + 2z_3 \quad (15)$$

If we choose these multiples so that

$$z_1 + z_2 + 3z_3 \leq 30 \quad (16)$$

$$2z_1 + 2z_2 + 5z_3 \leq 24 \quad (17)$$

$$4z_1 + z_2 + 2z_3 \leq 36 \quad (18)$$

we will have:

$$y_1(z_1 + z_2 + 3z_3) + y_2(2z_1 + 2z_2 + 5z_3) + y_3(4z_1 + z_2 + 2z_3) \leq 30y_1 + 24y_2 + 36y_3$$

Combining this with (15) we get

$$3z_1 + z_2 + 2z_3 \leq 30y_1 + 24y_2 + 36y_3$$

Consequently, finding the dual program $(P^*)^*$ of P^* amounts to maximizing the objective $3z_1 + z_2 + 2z_3$ subject to the constraints (16-18). But notice that,

except for having different variables, $(P^*)^*$ is exactly our starting program P . Thus, the dual program $(P^*)^*$ for program P^* is just P itself, i.e., $(P^*)^* = P$.

So, at the first sight, looking for the multipliers y_1, y_2, y_3 did not help much, because it only reduced a maximization problem to an equally hard minimization problem. Well, perhaps it still maybe easier to somehow solve both problems at the same, and this is precisely what the *SIMPLEX* algorithms does.²

To find the maximal value of $3x_1 + x_2 + 2x_3$ subject to the constraints

$$\begin{aligned}x_1 + x_2 + 3x_3 &\leq 30 \\2x_1 + 2x_2 + 5x_3 &\leq 24 \\4x_1 + x_2 + 2x_3 &\leq 36\end{aligned}$$

let us start with $x_1 = x_2 = x_3 = 0$ and ask ourselves: how much can we increase x_1 without violating the constraints? Since $x_1 + x_2 + 3x_3 \leq 30$ we can introduce a new variable x_4 such that $x_4 \geq 0$ and

$$x_4 = 30 - (x_1 + x_2 + 3x_3) \quad (19)$$

Since such variable measures how much “slack” we got between the actual value of $x_1 + x_2 + 3x_3$ and its upper limit 30, such x_4 is called a slack variable. Similarly we introduce new variables x_5, x_6 requiring them to satisfy $x_5, x_6 \geq 0$ and let

$$x_5 = 24 - 2x_1 - 2x_2 - 5x_3 \quad (20)$$

$$x_6 = 36 - 4x_1 - x_2 - 2x_3 \quad (21)$$

Since we started with the values $x_1 = x_2 = x_3 = 0$, this implies that these new slack variables must have values $x_4 = 30$, $x_5 = 24$, $x_6 = 36$, or as a single vector, the initial *basic feasible solution* is $(\overset{x_1}{0}, \overset{x_2}{0}, \overset{x_3}{0}, \overset{x_4}{30}, \overset{x_5}{24}, \overset{x_6}{36})$. Note that “feasible” refers merely to the fact that all of the constraints are satisfied.³

Now we see that (19) implies that x_1 cannot exceed 30, and (20) implies that $2x_1 \leq 24$, i.e., $x_1 \leq 12$, while (21) implies that $4x_1 \leq 36$, i.e., $x_1 \leq 9$. Since all of these conditions must be satisfied we conclude that x_1 cannot exceed 9, which is the upper limit coming from the constraint (21).

If we set $x_1 = 9$, this forces $x_6 = 0$. We now swap the roles of x_1 and x_6 : since we cannot increase x_1 any more, we eliminate x_1 from the right hand sides of the equations (19-21) and from the objective, introducing instead variable x_6 to the right hand side of the constraints and into the objective. To do that, we solve equation (21) for x_1 :

$$x_1 = 9 - \frac{x_2}{4} - \frac{x_3}{2} - \frac{x_6}{4}$$

²It is now useful to remember how we proved that the Ford - Fulkerson Max Flow algorithm in fact produces a maximal flow, by showing that it terminates only when the flow reaches the capacity of a (minimal) cut!

³Clearly, setting all variables to 0 does not always produce a basic feasible solution because this might violate some of the constraints; this would happen, for example, if we had a constraint of the form $-x_1 + x_2 + x_3 \leq -3$; choosing an initial basic feasible solution requires a separate algorithm to “bootstrap” the *SIMPLEX* algorithm - see for the details our (C-L-R-S) textbook.

and eliminate x_1 from the right hand side of the remaining constraints and the objective to get:

$$\begin{aligned}
z &= 3 \left(9 - \frac{x_2}{4} - \frac{x_3}{2} - \frac{x_6}{4} \right) + x_2 + 2x_3 \\
&= 27 - \frac{3}{4}x_2 - \frac{3}{2}x_3 - \frac{3}{4}x_6 + x_2 + 2x_3 \\
&= 27 + \frac{1}{4}x_2 + \frac{1}{2}x_3 - \frac{3}{4}x_6 \\
x_5 &= 24 - 2 \left(9 - \frac{x_2}{4} - \frac{x_3}{2} - \frac{x_6}{4} \right) - 2x_2 - 5x_3 \\
&= 6 + \frac{x_2}{2} + x_3 + \frac{x_6}{2} - 2x_2 - 5x_3 \\
&= 6 - \frac{3}{2}x_2 - 4x_3 + \frac{x_6}{2} \\
x_4 &= 30 - \left(9 - \frac{x_2}{4} - \frac{x_3}{2} - \frac{x_6}{4} \right) - x_2 - 3x_3 \\
&= 21 + \frac{x_2}{4} + \frac{x_3}{2} + \frac{x_6}{4} - x_2 - 3x_3 \\
&= 21 - \frac{3}{4}x_2 + \frac{5}{2}x_3 + \frac{x_6}{4}
\end{aligned}$$

To summarise: the “new” objective is

$$z = 27 + \frac{1}{4}x_2 + \frac{1}{2}x_3 - \frac{3}{4}x_6$$

and the “new constraints” are

$$x_1 = 9 - \frac{x_2}{4} - \frac{x_3}{2} - \frac{x_6}{4} \quad (22)$$

$$x_4 = 21 - \frac{3}{4}x_2 - \frac{5}{2}x_3 + \frac{x_6}{4} \quad (23)$$

$$x_5 = 6 - \frac{3}{2}x_2 - 4x_3 + \frac{x_6}{2} \quad (24)$$

Our new basic feasible solution replacing $(\overset{x_1}{0}, \overset{x_2}{0}, \overset{x_3}{0}, \overset{x_4}{30}, \overset{x_5}{24}, \overset{x_6}{36})$ is obtained by setting all the variables on the right to zero, thus obtaining $(\overset{x_1}{9}, \overset{x_2}{0}, \overset{x_3}{0}, \overset{x_4}{21}, \overset{x_5}{6}, \overset{x_6}{0})$.

NOTE: These are EQUIVALENT constraints and objectives; the old ones were only *transformed* to an equivalent form. Any values of the variables will produce exactly the same value in both forms of the objective and they will satisfy the first set of constraints if and only if they satisfy the second set.

So x_1 and x_6 have switched their roles; x_1 acts as a new basic variable, and the new basic feasible solution is: $(\overset{x_1}{9}, \overset{x_2}{0}, \overset{x_3}{0}, \overset{x_4}{21}, \overset{x_5}{6}, \overset{x_6}{0})$; the new value of the objective is $z = 27 + \frac{1}{4}0 + \frac{1}{2}0 - \frac{3}{4}0 = 27$. We will continue this process of finding basic feasible solutions which increase the value of the objective, switching which variables are used to measure the slack and which are on the right hand side of the constraint equations and in the objective. The variables on the left are called *the basic variables* and the variables on the right are the *non basic variables*.

We now choose another variable with a positive coefficient in the objective, say x_3 (we could also have chosen x_2). How much can we increase it?

From (22) we see that $\frac{x_3}{2}$ must not exceed 9, otherwise x_1 will become negative. Thus x_3 cannot be larger than 18. Similarly, $\frac{5}{2}x_3$ cannot exceed 21, otherwise x_4 will become negative, and so $x_3 \leq \frac{42}{5}$; similarly, $4x_3$ cannot exceed 6, ie, $x_3 \leq \frac{3}{2}$. Thus, in order for all constraints to remain valid, x_3 cannot exceed $\frac{3}{2}$. Thus, we increase x_3 to $\frac{3}{2}$; equation (24) now forces x_5 to zero. We now eliminate x_3 from the right hand side of the constraints and from the objective, taking it as a new basic variable:

$$4x_3 = 6 - \frac{3}{2}x_2 - x_5 + \frac{x_6}{2}$$

i.e.,

$$x_3 = \frac{3}{2} - \frac{3}{8}x_2 - \frac{1}{4}x_5 + \frac{1}{8}x_6 \quad (25)$$

After eliminating x_3 by substitution using (25), the objective now becomes:

$$\begin{aligned} z = 27 - \frac{1}{4}x_2 + \frac{1}{2} \left(\frac{3}{2} - \frac{3}{8}x_2 - \frac{1}{4}x_5 + \frac{1}{8}x_6 \right) - \frac{3}{4}x_6 = \\ \frac{111}{4} + \frac{1}{16}x_2 - \frac{1}{8}x_5 - \frac{11}{6}x_6 \end{aligned} \quad (26)$$

Using (25) to eliminate x_3 from the constraints, after simplifications we get the new constraints:

$$x_1 = \frac{33}{4} - \frac{x_2}{16} + \frac{x_5}{8} - \frac{5x_6}{16} \quad (27)$$

$$x_3 = \frac{3}{2} - \frac{3x_2}{8} - \frac{x_5}{4} + \frac{x_6}{8} \quad (28)$$

$$x_4 = \frac{69}{4} + \frac{3x_2}{16} + \frac{5x_3}{8} - \frac{x_6}{16} \quad (29)$$

Our new basic solution is again obtained using the fact that all variables on the right and in the objective, including the newly introduced non-basic variable x_5 , are equal to zero, i.e., the new basic feasible solution is $(\frac{33}{4}, 0, \frac{3}{2}, \frac{69}{4}, 0, 0)$ and the new value of the objective is $z = \frac{111}{4}$.

Comparing this with the previous basic feasible solution $(9, 0, 0, 21, 6, 0)$ we see that in the new basic feasible solution the value of x_1 has dropped from 9 to $33/4$; however, this now has no effect on the value of the objective, because x_1 no longer appears in the objective; all the variables appearing in the objective (thus, the *non-basic* variables) always have value 0.

We now see that the only variable in the objective (26) appearing with a positive coefficient is x_2 . How much can we increase it without violating the new constraints? The first constraint (27) implies that $\frac{x_2}{16} \leq \frac{33}{4}$, i.e., that $x_2 \leq 132$; the second constraint (28) implies that $\frac{3x_2}{8} \leq \frac{3}{2}$, i.e., that $x_2 \leq 4$. Note that the third constraint (29) does not impose any restrictions on how large x_2 can be, for as long as it is positive; thus, we conclude that the largest possible value

of x_2 which does not cause violation of any of the constraints is $x_2 = 4$, which corresponds to constraint (28). The value $x_2 = 4$ forces $x_3 = 0$; we now switch the roles of x_2 and x_3 (this operation of switching the roles of two variables is called *pivoting*) by solving (28) for x_2 :

$$x_2 = 4 - \frac{8x_3}{3} - \frac{2x_5}{3} + \frac{x_6}{3}$$

and then using this to eliminate x_2 from the objective, obtaining

$$z = 28 - \frac{x_3}{6} - \frac{x_5}{6} - \frac{2x_6}{3}$$

as well as from the constraints, obtaining after simplification

$$x_1 = 8 + \frac{x_3}{6} + \frac{x_5}{6} - \frac{x_6}{3} \tag{30}$$

$$x_2 = 4 - \frac{8x_3}{3} - \frac{2x_5}{3} + \frac{x_6}{3} \tag{31}$$

$$x_4 = 18 - \frac{x_3}{2} + \frac{x_5}{2} \tag{32}$$

which produces the new basic feasible solution $(\overset{x_1}{8}, \overset{x_2}{4}, \overset{x_3}{0}, \overset{x_4}{18}, \overset{x_5}{0}, \overset{x_6}{0})$ with the new value of the objective $z = 28$. Note that in the new objective all the variables appear with a negative coefficient; thus our procedure terminates, but did it find the maximum value of the objective? Maybe with a different choices of variables in pivoting we would have come up with another basic feasible solution which would have different basic variables, also with all non basic variables in the objective appearing with a negative coefficient, but for which the obtained value of the objective is larger?

This is not the case: just as in the case of the Ford Fulkerson algorithm for Max Flow, once the pivoting terminates, the solution must be optimal regardless of which particular variables were swapped in pivoting, because **the pivoting terminates when the corresponding basic feasible solution of the program becomes equal to a basic feasible solution of the dual program**. Since *every* feasible solution of the dual is larger than every feasible solution of the starting (or *primal*) program, we get that the SIMPLEX algorithm must return the optimal value after it terminates. We now explain this in more detail.

1 LP Duality

General setup

Comparing our initial program P with its dual P^* :

$$\begin{array}{ll}
 P : \text{ maximize} & 3x_1 + x_2 + 2x_3, \\
 \text{subject to the constraints} & \\
 & x_1 + x_2 + 3x_3 \leq 30 \\
 & 2x_1 + 2x_3 + 5x_3 \leq 24 \\
 & 4x_1 + x_2 + 2x_3 \leq 36 \\
 & x_1, x_2, x_3 \geq 0;
 \end{array}$$

$$\begin{array}{ll}
 P^* : \text{ minimize} & 30y_1 + 24y_2 + 36y_3, \\
 \text{subject to the constraints} & \\
 & y_1 + 2y_2 + 4y_3 \geq 3 \\
 & y_1 + 2y_2 + y_3 \geq 1 \\
 & 3y_1 + 5y_2 + 2y_3 \geq 2 \\
 & y_1, y_2, y_3 \geq 0.
 \end{array}$$

we see that the original, *primal* Linear Program P and its *dual* Linear Program are related as follows

$$\begin{array}{ll}
 P : \text{ maximize} & z(x) = \sum_{j=1}^n c_j x_j, \\
 \text{subject to the constraints} & \sum_{j=1}^n a_{ij} x_j \leq b_i; \quad 1 \leq i \leq m \\
 & x_1, \dots, x_n \geq 0;
 \end{array}$$

$$\begin{array}{ll}
 P^* : \text{ minimize} & z^*(y) = \sum_{i=1}^m b_i y_i, \\
 \text{subject to the constraints} & \sum_{i=1}^m a_{ij} y_i \geq c_j; \quad 1 \leq j \leq n \\
 & y_1, \dots, y_m \geq 0,
 \end{array}$$

or, in matrix form,

$$\begin{array}{ll}
 P : \text{ maximize} & z(x) = c^T x, \text{ subject to the constraints} \quad Ax \leq b \text{ and } x \geq 0; \\
 P^* : \text{ minimize} & z^*(y) = b^T y, \text{ subject to the constraints} \quad A^T y \geq c \text{ and } y \geq 0.
 \end{array}$$

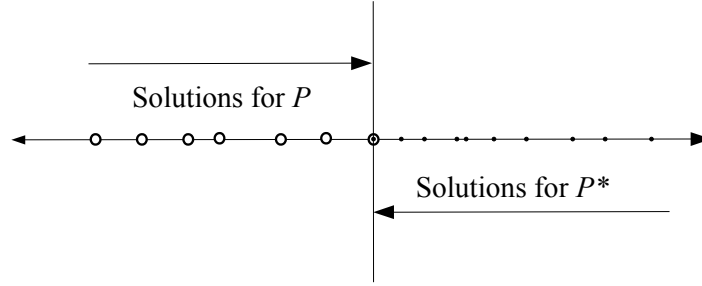
Weak Duality Theorem If $x^\top = \langle x_1, \dots, x_n \rangle$ is any basic feasible solution for P and $y^\top = \langle y_1, \dots, y_m \rangle$ is any basic feasible solution for P^* , then:

$$z(x) = \sum_{j=1}^n c_j x_j \leq \sum_{i=1}^m b_i y_i = z^*(y)$$

Proof: Since x and y are basic feasible solutions for P and P^* respectively, we can use the constraint inequalities, first from P^* and then from P to obtain

$$z(x) = \sum_{j=1}^n c_j x_j \leq \sum_{j=1}^n \left(\sum_{i=1}^m a_{ij} y_i \right) x_j = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} x_j \right) y_i \leq \sum_{i=1}^m b_i y_i = z^*(y)$$

Thus, every feasible solution of P^* is an upper bound for the set of all feasible solutions of P , and every feasible solution of P is a lower bound for the set of feasible solutions for P^* ; see the figure below.



Consequently, if we find a feasible solution of P which is also a feasible solution for P^* , such solution must be maximal feasible solution for P and minimal feasible solution for P^* .

We now show that when the *SIMPLEX* algorithm terminates that it produces a basic feasible solution $\bar{x}$ for P , and, implicitly, a basic feasible solution $\bar{y}$ for the dual P^* for which $z(\bar{x}) = z^*(\bar{y})$; by the above, this will imply that $z(\bar{x})$ is the maximal value for the objective of P and that $z^*(\bar{y})$ is the minimal value of the objective for P^* .

Assume that the *SIMPLEX* algorithm has terminated; let B be such that the basic variables (variables on the left hand side of the constraint equations) in the final form of P are variables x_i for which $i \in B$; let $N = \{1, 2, \dots, n+m\} \setminus B$; then x_j for $j \in N$ are all the non-basic variables in the final form of P . Since the *SIMPLEX* algorithm has terminated, we have also obtained a set of coefficients $\bar{c}_j \leq 0$ for $j \in N$, as well as $\bar{v}$ such that the final form of the objective is

$$z(x) = \bar{v} + \sum_{j \in N} \bar{c}_j x_j$$

If we set all the final non-basic variables x_j , $j \in N$, to zero, we obtain a basic feasible solution $\bar{x}$ for which $z(\bar{x}) = \bar{v}$.

Let us define $\bar{c}_j = 0$ for all $j \in B$; then

$$\begin{aligned}
z(x) &= \sum_{j=1}^n c_j x_j \\
&= \bar{v} + \sum_{j=1}^{n+m} \bar{c}_j x_j \\
&= \bar{v} + \sum_{j=1}^n \bar{c}_j x_j + \sum_{i=n+1}^{n+m} \bar{c}_i x_i \\
&= \bar{v} + \sum_{j=1}^n \bar{c}_j x_j + \sum_{i=1}^m \bar{c}_{n+i} x_{n+i}
\end{aligned}$$

Since the variables x_{n+i} , ($1 \leq i \leq m$), are the initial slack variables, they satisfy $x_{n+i} = b_i - \sum_{j=1}^n a_{ij} x_j$; thus we get

$$\begin{aligned}
z(x) &= \sum_{j=1}^n c_j x_j \\
&= \bar{v} + \sum_{j=1}^n \bar{c}_j x_j + \sum_{i=1}^m \bar{c}_{n+i} \left(b_i - \sum_{j=1}^n a_{ij} x_j \right) \\
&= \bar{v} + \sum_{j=1}^n \bar{c}_j x_j + \sum_{i=1}^m \bar{c}_{n+i} b_i - \sum_{i=1}^m \sum_{j=1}^n \bar{c}_{n+i} a_{ij} x_j \\
&= \bar{v} + \sum_{j=1}^n \bar{c}_j x_j + \sum_{i=1}^m \bar{c}_{n+i} b_i - \sum_{j=1}^n \sum_{i=1}^m \bar{c}_{n+i} a_{ij} x_j \\
&= \bar{v} + \sum_{i=1}^m \bar{c}_{n+i} b_i + \sum_{j=1}^n \left(\bar{c}_j - \sum_{i=1}^m \bar{c}_{n+i} a_{ij} \right) x_j
\end{aligned}$$

The above equations hold true for all values of x ; thus, comparing the first and the last equation we conclude that

$$\begin{aligned}
\bar{v} + \sum_{i=1}^m \bar{c}_{n+i} b_i &= 0; \\
\bar{c}_j - \sum_{i=1}^m \bar{c}_{n+i} a_{ij} &= c_j, \quad (1 \leq j \leq n).
\end{aligned}$$

i.e.,

$$\begin{aligned}
\sum_{i=1}^m b_i (-\bar{c}_{n+i}) &= \bar{v}; \\
\sum_{i=1}^m a_{ij} (-\bar{c}_{n+i}) &= c_j + (-\bar{c}_j), \quad (1 \leq j \leq n).
\end{aligned}$$

We now see that if we set $\bar{y}_i = -\bar{c}_{n+i}$ for all $1 \leq i \leq m$, then, since the *SIMPLEX* terminates when all coefficients of the objective are either negative or zero, then such $\bar{y}$ satisfies:

$$\sum_{i=1}^m b_i \bar{y}_i = \bar{v};$$

$$\sum_{i=1}^m a_{ij} \bar{y}_i = c_j - \bar{c}_j \geq c_j, \quad (1 \leq j \leq n),$$

$$\bar{y}_i \geq 0, \quad (1 \leq i \leq m).$$

P : (initial form)	initial objective: $z(x) = \sum_{j=1}^n c_j x_j$ $c_1 \quad c_2 \quad c_3 \quad c_4 \quad \dots \quad c_{n-3} \quad c_{n-2} \quad c_{n-1} \quad c_n$ $(x_1) \quad (x_2) \quad (x_3) \quad (x_4) \quad \dots \quad (x_{n-3}) \quad (x_{n-2}) \quad (x_{n-1}) \quad (x_n)$ initial non-basic variables	$x_{n+1} \quad x_{n+2} \quad x_{n+3} \quad \dots \quad x_{n+m-2} \quad x_{n+m-1} \quad x_{n+m}$ initial slack variables
initial basic feasible solution	0 0 0 0 ... 0 0 0 0	
	•	•
	•	•
	•	•

P : (after termination)	final objective: $z(x) = \sum_{j=1}^{n+m} \bar{c}_j x_j; \quad \bar{c}_j = 0 \text{ for } j \in B$ $\bar{c}_1 \quad 0 \quad \bar{c}_3 \quad 0 \quad \dots \quad 0 \quad \bar{c}_{n-2} \quad 0 \quad \bar{c}_n \quad 0 \quad \bar{c}_{n+2} \quad \bar{c}_{n+3} \quad \dots \quad 0 \quad 0 \quad \bar{c}_{n+m}$ $(x_1) \quad x_2 \quad (x_3) \quad x_4 \quad \dots \quad x_{n-3} \quad (x_{n-2}) \quad x_{n-1} \quad (x_n)$ (final NON-BASIC variables $x_j, j \in B$ circled)	$x_{n+1} \quad (x_{n+2}) \quad (x_{n+3}) \quad \dots \quad x_{n+m-2} \quad x_{n+m-1} \quad (x_{n+m})$
final basic feasible solution	0 $\bar{x}_2$ 0 $\bar{x}_4$... $\bar{x}_3$ 0 $\bar{x}_{n-1}$ 0	$\bar{x}_{n+1}$ 0 0 ... $\bar{x}_{n+m-2}$ $\bar{x}_{n+m-1}$ 0

Dual P^* : (mixed form)	dual slack variables $y_{m+j} - y_{n+m}$ $(y_{m+1}) \quad y_{m+2} \quad (y_{m+3}) \quad y_{m+4} \quad \dots \quad y_{n+m-3} \quad (y_{n+m-2}) \quad y_{n-1} \quad (y_{n+m})$ (final BASIC dual variables circled)	initial dual objective: $z^*(x) = \sum_{j=1}^m b_j y_j;$ $y_1 \quad (y_2) \quad (y_3) \quad \dots \quad y_2 \quad y_1 \quad (y_m)$
dual basic feasible solution	$-\bar{c}_1 \quad 0 \quad -\bar{c}_3 \quad 0 \quad \dots \quad 0 \quad \bar{c}_{n-2} \quad 0 \quad \bar{c}_n$	0 $-\bar{c}_{n+2}$ $-\bar{c}_{n+3}$ 0 0 $-\bar{c}_{n+m}$

For all final non-basic variables of the form x_{n+j} (in the right box and circled) the corresponding basic dual variable y_j is set to $-\bar{c}_{n+j}$; for all final basic variables of the form x_{n+j} (in the right box and NOT circled) the value of y_j is set to 0.

Thus, such $\bar{y}$ is a basic feasible solution for the dual program P^* for which the dual objective has the same value $\bar{v}$ which the original, primal program P achieves for the basic feasible solution $\bar{x}$. Thus, by the Weak Duality Theorem, we conclude that $\bar{v}$ is the maximal feasible value of P and minimal feasible value for P^* .

Note also that the basic and non basic variables of the final primal form of the problem and of its dual are complementary: for every $1 \leq i \leq m$, variable x_{n+i} is basic for the final form of the primal if and only if y_i is non basic for the final form of the dual; (similarly, for every $1 \leq j \leq n$, variable x_j is basic for the final form of the primal if and only if y_{m+j} is not basic for the final form of the dual). Since the basic variables measure the slack of the corresponding basic feasible solution, we get that if $\bar{x}$ and $\bar{y}$ are the extremal feasible solutions

for P and P^* , respectively, then for all $1 \leq j \leq n$ and all $1 \leq i \leq m$,

$$\begin{aligned} \text{either } \bar{x}_j = 0 \quad \text{or} \quad y_{m+j} = 0, \quad \text{i.e.,} \quad \sum_{i=0}^m a_{ij} \bar{y}_i = c_j; \\ \text{either } \bar{y}_i = 0 \quad \text{or} \quad x_{n+i} = 0, \quad \text{i.e.,} \quad \sum_{j=0}^n a_{ij} \bar{x}_j = b_i. \end{aligned}$$

Note that any equivalent form of P which is obtained through a pivoting operation is uniquely determined by its corresponding set of basic variables. Assuming that we have n variables and m equations, then there are $\binom{n+m}{m}$ choices for the set of basic variables. Using the Stirling formula

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \Rightarrow \ln n! \approx \frac{\ln(2\pi n)}{2} + n \ln n - n = n \ln n - n + O(\ln n)$$

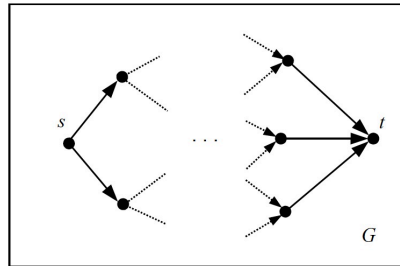
we get

$$\begin{aligned} \ln \binom{n+m}{m} &= \ln \frac{(m+n)!}{m! n!} = \ln(m+n)! - \ln m! - \ln n! \\ &= (m+n) \ln(m+n) - (m+n) - n \ln n - m \ln m + m + n \\ &= m(\ln(m+n) - \ln m) + n(\ln(m+n) - \ln n) \\ &\geq m + n \end{aligned}$$

Thus, the total number of choices for the set of the basic variables is $\binom{n+m}{m} > e^{m+n}$. This implies that the *SIMPLEX* algorithm could potentially run in exponential time, and in fact, one can construct examples of LP on which the run time of the *SIMPLEX* algorithm is exponential. However, in practice the *SIMPLEX* algorithm is extremely efficient, even for large problems with thousands of variables and constraints, and it tends to outperform algorithms for LP which do run in polynomial time (the Ellipsoid Method and the Interior Points Method).

1.1 Examples of dual programs: max flow

We would now like to formulate the Max Flow problem in a flow network as a Linear Program.

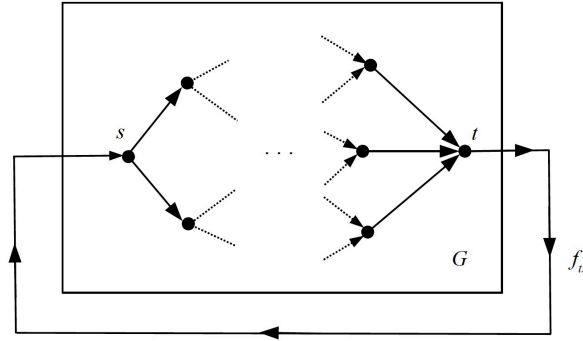


Thus, assume we are given a flow network G with capacities κ_{ij} of all edges $(i, j) \in G$. The max flow problem seeks to maximize the flows through a network

flow graph G , subject to the constraints:

$$C^* \begin{cases} f_{ij} & \leq \kappa_{ij}; \quad (i, j) \in G; \quad (\text{flow smaller than pipe's capacity}) \\ \sum_{i: (i,j) \in G} f_{ij} & = \sum_{k: (j,k) \in G} f_{jk}; \quad j \in G; \quad (\text{incoming flow equals outgoing}) \\ f_{ij} & \geq 0; \quad (i, j) \in G \quad (\text{no negative flows}). \end{cases}$$

To eliminate the equality in the second constraint in C^* while introducing only one rather than two inequalities, we use a “trick”: we make the flow circular, by connecting the sink t with the source s with a pipe of infinite capacity. Thus, we now have a new graph G' , $G \subset G'$, with an additional edge $(t, s) \in G'$ with capacity ∞ .



We can now formulate the Max Flow problem as a Linear Program by replacing the equality in the second constraint with a single but equivalent inequality:

P: *maximize:* f_{ts}
subject to the constraints:

$$\begin{aligned} f_{ij} &\leq \kappa_{ij}; & (i, j) \in G; \\ \sum_{i: (i,j) \in G'} f_{ij} - \sum_{k: (j,k) \in G'} f_{jk} &\leq 0; & j \in G; \\ f_{ij} &\geq 0; & (i, j) \in G'. \end{aligned}$$

Thus, the coefficients c_{ij} of the primal P are zero for all variables f_{ij} except for f_{ts} which is equal to 1, i.e.,

$$z(f) = \sum_{ij} 0 \cdot f_{ij} + 1 \cdot f_{ts} \quad (1.1)$$

To obtain the dual of P we look for coefficients d_{ij} , $(i, j) \in G$ corresponding to the first set of constraints, and coefficients p_j , $j \in G$ corresponding to the second set of constraints to use as multipliers of the constraints:

$$\begin{aligned} f_{ij} d_{ij} &\leq \kappa_{ij} d_{ij}; & (i, j) \in G; \\ \sum_{i: (i,j) \in G'} f_{ij} p_j - \sum_{k: (j,k) \in G'} f_{jk} p_j &\leq 0; & j \in G. \end{aligned}$$

note the two special cases of the second inequality involving f_{ts} are

$$\begin{aligned} \sum_{i: (i,t) \in G} f_{it}p_t - f_{ts}p_t &\leq 0; \\ f_{ts}p_s - \sum_{k: (s,k) \in G} f_{sk}p_s &\leq 0. \end{aligned}$$

Summing these inequalities and factoring out, we get

$$\sum_{(i,j) \in G} (d_{ij} - p_i + p_j)f_{ij} + (p_s - p_t)f_{ts} \leq \sum_{(i,j) \in G} \kappa_{ij}d_{ij}$$

Thus, the dual objective is the right hand side of the above inequality, and, as before, the dual constraints are obtained by comparing the coefficients of the left hand side with the coefficients of the objective:

$$P^* : \text{ minimize: } \sum_{(i,j) \in G} \kappa_{ij}d_{ij}$$

subject to the constraints:

$$\begin{aligned} d_{ij} - p_i + p_j &\geq 0 & (i,j) \in G \\ p_s - p_t &\geq 1 \end{aligned}$$

$$\begin{aligned} d_{ij} &\geq 0 & (i,j) \in G \\ p_i &\geq 0 & i \in G \end{aligned}$$

Let us write the constraints of P using new slack variables ψ_{ij}, φ_j :

$$\begin{aligned} \psi_{ij} &= \kappa_{ij} - f_{ij}; & (i,j) \in G; \\ \varphi_j &= \sum_{k: (j,k) \in G'} f_{jk} - \sum_{i: (i,j) \in G'} f_{ij}; & j \in G; \\ f_{ij} &\geq 0; & (i,j) \in G'; \\ \psi_{ij} &\geq 0; & (i,j) \in G'; \\ \varphi_j &\geq 0; & j \in G \end{aligned}$$

Note now an important feature of both P and P^* : all variables $f_{ij}, \psi_{ij}, \varphi_j$ in P (and also all variables d_{ij}, p_i in P^*) appear only with the coefficient ± 1 . One can now see that, if we solve the above constraint equations for any subset of the set of all variables $\{f_{ij}, \psi_{ij}, \varphi_j : (i,j) \in G^*, j \in G\}$ as the set of the basic variables, all the coefficients c_{ij} in the new objective which multiply ψ_{ij} or φ_j and which are obtained after the corresponding substitutions removing the basic variables from the objective will have coefficients either 0 or 1.¹ This means that in the corresponding basic feasible solution of P' at which the minimal value of the dual program is obtained, also all values $\bar{d}_{ij}$ of d_{ij} and all values $\bar{p}_j$ of p_j will be either 0 or 1.

What is the interpretation of such solution of the dual Linear Program P^* ? Let us consider the set A of all vertices j of G for which $\bar{p}_j = 1$ and the set B of

¹This corresponds to the fact that such constraints define a polyhedra whose all vertices have coordinates which are all either 0 or 1.

all vertices j for which $\bar{p}_j = 0$. Then $A \cup B = G$ and $A \cap B = \emptyset$. The constraint $p_s - p_t \geq 1$ of P^* implies that $\bar{p}_s = 1$ and $\bar{p}_t = 0$, i.e., $s \in A$ and $t \in B$. Thus, A and B define a cut in the flow network. Since at points $\bar{p}, \bar{d}$ the objective $\sum_{(i,j) \in G} \kappa_{ij} d_{ij}$ achieves the minimal value, the constraint $d_{ij} - p_i + p_j \geq 0$ implies that $\bar{d}_{ij} = 1$ if and only if $p_i = 1$ and $p_j = 0$, i.e., if and only if the edge (i, j) has crossed from set A into set B . Thus, the minimum value of the dual objective $\sum_{(i,j) \in G} \kappa_{ij} d_{ij}$ precisely corresponds to the capacity of the cut defined by A, B . Since such value is equal to the maximal value of the flow defined by the primal problem, we have obtained a maximal flow and minimal cut in G !

As we have mentioned, the extreme values of linear optimization problems are always obtained on the vertices of the corresponding constraint polyhedra. In *this particular case* all vertices are with $0, 1$ coordinates; however, in general this is *false*. For example, NP hard problems formulated as Linear Programming problems always result in polyhedra with non-integer vertices.

Theorem: Solving an Integer Linear Program (ILP), i.e., an LP with additional constraints that the values of all variables must be integers is NP hard, i.e., there cannot be a polynomial time algorithm for solving *ILPs* (unless an extremely unlikely thing happens, namely that $P = NP$).

Extending the scope of LP. There are several types of problems which are not linear programs per se, but which can be reduced to linear programs by clever tricks. We give two such examples below.

Example: Assume that you are given a set of 10 points in the plane, with coordinates (x_i, y_i) , $1 \leq i \leq 10$. Your goal is to find a polynomial of degree 3 which "fits" the data as closely as possible. What does this mean? In order for the requirement to make sense we must specify how we measure how close such polynomial is to the given data. Let us represent the polynomial in the form

$$P(x) = a_3 x^3 + a_2 x^2 + a_1 x + a_0$$

where a_3, a_2, a_1, a_0 are the coefficients to be determined.

1. **In the sense of the L^2 norm:** We need to minimize the sum

$$S_2(a_0, a_1, a_2, a_3) = \sum_{i=1}^{10} (P(x_i) - y_i)^2 = \sum_{i=1}^{10} (a_3 x_i^3 + a_2 x_i^2 + a_1 x_i + a_0 - y_i)^2$$

which means that we want to minimize the sum of the squares of the distances between the values of $P(x)$ at points $x_1, \dots, x_{10}$ and the points with coordinates y_i . This is a typical "Least Squares" approximation, and it is solved by computing the partial derivatives of $S(a_0, a_1, a_2, a_3)$ and setting them equal to 0:

$$\frac{\partial S_2(a_0, a_1, a_2, a_3)}{\partial a_k} = \sum_{i=1}^{10} (a_3 x_i^3 + a_2 x_i^2 + a_1 x_i + a_0 - y_i) x_i^k = 0$$

Note that this results in a system of linear equations in unknowns a_0, a_1, a_2, a_3 which is easy to solve.

2. **In the sense of the uniform norm:** We need to minimize

$$\begin{aligned} S_U(a_0, a_1, a_2, a_3) &= \max \{|P(x_i) - y_i|, 1 \leq i \leq 10\} \\ &= \max \{|a_3x_i^3 + a_2x_i^2 + a_1x_i + a_0 - y_i|, 1 \leq i \leq 10\} \end{aligned}$$

Note that, as it stands, this is nNOT an LP problem (or program, as it is called) because it involves two non-linear operations: taking the max and taking absolute values. However, we can reduce it to an LP by introducing a new variable u and solving the following problem:

$$\begin{array}{ll} \text{minimize} & u \\ \text{subject to the constraints} & |P(x_i) - y_i| \leq u, \quad 1 \leq i \leq 10 \end{array}$$

Note that in the above minimization problem the objective is extremely simple, it is just the variable u which plays the role of an upper bound for all of the absolute values of the differences $|P(x_i) - y_i| \leq u$, $1 \leq i \leq 10$. This way we got rid of the non-linear max operator, but now we also have to get rid of the absolute values of all the differences. To do that, we note that, if $u > 0$, then $|z| \leq u$ if and only if both $z \leq u$ and $-z \leq u$. Thus, we solve the following minimization problem:

$$\begin{array}{ll} \text{minimize} & u \\ \text{subject to the constraints} & P(x_i) - y_i \leq u \ \& \ -(P(x_i) - y_i) \leq u \quad 1 \leq i \leq 10 \end{array}$$

which is equivalent to:

$$\begin{array}{ll} \text{minimize} & u \\ \text{subject to the constraints} & \begin{aligned} a_3x_i^3 + a_2x_i^2 + a_1x_i + a_0 - u &\leq y_i & 1 \leq i \leq 10 \\ -(a_3x_i^3 + a_2x_i^2 + a_1x_i + a_0) - u &\leq -y_i & 1 \leq i \leq 10 \end{aligned} \end{array}$$

This is an LP almost in the standard form: we now only add inequality $u \geq 0$, but, since the coefficients a_i can be both positive and negative, we now do the trick we have already mentioned: we replace each a_i with $\bar{a}_i - \hat{a}_i$, where $\bar{a}$ and $\hat{a}$ are two new variables. Thus our polynomial now is $(\bar{a}_3 - \hat{a}_3)x^3 + (\bar{a}_2 - \hat{a}_2)x^2 + (\bar{a}_1 - \hat{a}_1)x + (\bar{a}_0 - \hat{a}_0)$, and we can add the constraints $\bar{a}_i \geq 0$ and $\hat{a}_i \geq 0$, thus obtaining a proper standard form LP.

3. **In the sense of the L^1 norm:** We need to minimize the sum

$$S_1(a_0, a_1, a_2, a_3) = \sum_{i=1}^{10} |P(x_i) - y_i| = \sum_{i=1}^{10} |a_3x_i^3 + a_2x_i^2 + a_1x_i + a_0 - y_i|$$

This can be reduced to an LP program by a somewhat similar trick: for every i such that $1 \leq i \leq 10$ we introduce new variables u_i and solve the

following minimization problem:

$$\begin{aligned}
& \text{minimize} && \sum_{i=1}^{10} u_i \\
& \text{subject to the constraints} && P(x_i) - y_i \leq u_i \ \& \ - (P(x_i) - y_i) \leq u_i \quad 1 \leq i \leq 10
\end{aligned}$$

which is equivalent to:

$$\begin{aligned}
& \text{minimize} && \sum_{i=1}^{10} u_i \\
& \text{subject to the constraints} && a_3 x_i^3 + a_2 x_i^2 + a_1 x_i + a_0 - u_i \leq y_i \quad 1 \leq i \leq 10 \\
& && - (a_3 x_i^3 + a_2 x_i^2 + a_1 x_i + a_0) - u_i \leq -y_i \quad 1 \leq i \leq 10
\end{aligned}$$

which is again reducible to an LP program by replacing each a_i with $\bar{a}_i - \hat{a}_i$, just as in the previous case.